

Nut consumption and blood lipid levels: a pooled analysis of 25 intervention trials.

BACKGROUND: Epidemiological studies have consistently associated nut consumption with reduced risk for coronary heart disease. Subsequently, many dietary intervention trials investigated the effects of nut consumption on blood lipid levels. The objectives of this study were to estimate the effects of nut consumption on blood lipid levels and to examine whether different factors modify the effects. **METHODS:** We pooled individual primary data from 25 nut consumption trials conducted in 7 countries among 583 men and women with normolipidemia and hypercholesterolemia who were not taking lipid-lowering medications. In a pooled analysis, we used mixed linear models to assess the effects of nut consumption and the potential interactions. **RESULTS:** With a mean daily consumption of 67 g of nuts, the following estimated mean reductions were achieved: total cholesterol concentration (10.9 mg/dL [5.1% change]), low-density lipoprotein cholesterol concentration (LDL-C) (10.2 mg/dL [7.4% change]), ratio of LDL-C to high-density lipoprotein cholesterol concentration (HDL-C) (0.22 [8.3% change]), and ratio of total cholesterol concentration to HDL-C (0.24 [5.6% change]) ($P < .001$ for all) (to convert all cholesterol concentrations to millimoles per liter, multiply by 0.0259). Triglyceride levels were reduced by 20.6 mg/dL (10.2%) in subjects with blood triglyceride levels of at least 150 mg/dL ($P < .05$) but not in those with lower levels (to convert triglyceride level to millimoles per liter, multiply by 0.0113). The effects of nut consumption were dose related, and different types of nuts had similar effects on blood lipid levels. The effects of nut consumption were significantly modified by LDL-C, body mass index, and diet type: the lipid-lowering effects of nut consumption were greatest among subjects with high baseline LDL-C and with low body mass index and among those consuming Western diets. **CONCLUSION:** Nut consumption improves blood lipid levels in a dose-related manner, particularly among subjects with higher LDL-C or with lower BMI.